

Pre-merge review of the iclr2027-eval-plan branch

bkrobust — ICLR 2027 submission

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11 September 2026

1 Summary

The branch is **purely additive**: nine new files, **2,169 insertions**, **zero deletions**, and **no change to any file under src/**. It contributes one planning document, two read-only analysis scripts, the four result files those scripts emit, a drafted Methods paragraph, and a three-line `.gitignore` amendment that un-ignores the new results directory.

The contribution is best understood not as new machinery but as a **forensic audit of the corpus the project already had committed**. Its central result is negative and consequential: the existing oracle corpus cannot, even in principle, exhibit variation in the quantity the paper intends to headline. The plan then re-scopes the evaluation around that finding.

I reproduced the committed artifacts and re-derived the principal claims independently. Everything I could check held. The issues listed in Section 6 are documentation and consistency defects, not computational errors.

Recommendation: **merge**, with the four documentation corrections in Section 6 applied before the 18 September abstract deadline.

2 What changed

File	Kind	Lines	Role
docs/EVALUATION_PLAN.md	doc	158	The plan and its evidence
experiments/stage0_ladder.py	script	194	Free-rule scoring
experiments/stage0_claim_radius.py	script	202	Claim-unit radius, 3 checks
paper/sections/evaluation-protocol.tex	paper	27	Methods paragraph (draft)
results/stage0/ladder_rows.csv	result	832	831 scored rows
results/stage0/ladder_summary.json	result	112	Aggregates
results/stage0/claim_radius_rows.csv	result	541	540 retraction rows
results/stage0/claim_radius_summary.json	result	100	Aggregates
.gitignore	config	3	Un-ignore results/stage0/
Total		2,169	(0 deletions)

Both scripts read `results/axisa3/instances.jsonl` and write only into `results/stage0/`. Neither mutates the pipeline. `stage0_ladder.py` imports nothing from `bkrobust`; `stage0_claim_radius.py` imports seven functions and calls no writer.

3 Methodology

3.1 The input corpus

results/axisa3/instances.jsonl holds **116,094 records**. Of these, 120 are `_meta` header records and 115,143 are rejections, leaving **831 admissible rows** — an admissibility rate of **0.72%**. The rejection funnel, which is the substance of the plan’s proposed Table 1:

Disposition	Rows	Meaning
treatment_not_in_or_adjacent_to_component	66,330	No undirected structure at X
no_causal_path	42,618	$Y \notin \text{de}(X)$
empty_set_trivially_valid	3,738	Nothing to adjust for
no_atomic_perturbation_changes_validity	1,659	<i>Claim radius</i> ≥ 2
o_g0_not_identified	518	Optimal set undefined at G_0
o_g0_extensions_intractable	280	Measurement limit, not structure
<code>_meta</code> records	120	39 networks $\times$ 3 coverages, +3
admissible	831	543 distinct (<i>network</i> , X , Y) pairs

The 831 rows split by coverage as 543 / 182 / 106 at coverage 1.0 / 0.5 / 0.25, and by dispatch as 811 rows via `local_up_fast` and 20 via `e1_ladder`.

3.2 stage0_ladder.py — scoring a free heuristic

The script defines the *free rule*: predict the graph separation s (the distance inside X ’s chain component to the nearest member of the adjustment set) where it is measured, and predict 1 where it is undefined. It then scores that prediction against the committed radius on all 831 rows, and — importantly — **signs every miss**. A prediction above the committed radius *over-certifies* (it claims more safety than exists, and is the only dangerous direction); a prediction below it is merely conservative.

It additionally computes: the direction of the prior “session-5 law” on the 463 rows where separation is measured; the pairing of instances across the three coverage levels; and a one-way ANOVA intraclass correlation of the $r = 1$ indicator, grouped by network, to establish the effective sample size.

3.3 stage0_claim_radius.py — the radius in the analyst’s own units

This is the substantive script. It performs three checks.

Check 1, single-claim retraction. For each of the 540 coverage-1.0 rows (excluding `pathfinder`), it takes the recovering knowledge set K , retracts each asserted claim in turn, re-closes the CPDAG under Meek’s rules, and tests whether the committed optimal adjustment set is still GAC-valid in the resulting MPDAG. A row broken by *some* single retraction has a claim radius of 1, whatever its hop-count radius says. It also records ϕ_1 , the *fraction* of single retractions that break validity.

Check 2, nesting. Whether $K(0.25) \subseteq K(0.5) \subseteq K(1.0)$ under the stride that `benchmarks.measure.select_knowledge` uses.

Check 3, amenability. Whether the 518 `o_g0_not_identified` rejections are amenable at G_0 . A non-amenable query has no valid adjustment set at all, so the rejection is structural rather than an artefact of how the optimal set was defined.

4 Findings

4.1 F1. The claim radius is 1 on every row — and this is forced, not measured

All **540 of 540** coverage-1.0 rows are broken by a single claim retraction. The committed *hop* radius on those same rows is 1 in 377, 2 in 102, 3 in 43, and 4–14 in 18. Median asserted set size is 5 claims; median closure size k_{G_0} is 12.

The colleague correctly identifies that this is definitional rather than empirical, and I confirmed the mechanism at its source. `fast_gate` (`src/bkrobust/benchmarks/measure.py:62`) admits a pair precisely when some single retraction from k_{true} breaks GAC-validity of the optimal set. At coverage 1.0 the asserted set *is* the recovering set and G_0 *is* the true DAG, so the gate’s admission test and Check 1 are the same test. The result is a tautology, and the plan says so.

[flag] An implication worth stating explicitly, which the plan leaves implicit. The complement is visible in the funnel: the **1,659** rows rejected as `no_atomic_perturbation_changes_validity` are *exactly* the population with claim radius ≥ 2 . The gate does not merely fail to vary the headline quantity — it discards, before measurement, the only rows that could have varied it. Any attempt to calibrate the claim radius on the committed corpus is therefore not just underpowered but structurally impossible. This strengthens the plan’s conclusion that the elicited arm must carry the paper’s numbers; it also suggests a cheap second option the plan does not mention, namely re-running the sweep with the gate’s atomic-perturbation clause recorded as a status column instead of applied as a filter, which would recover those 1,659 rows as the natural radius- ≥ 2 stratum.

ϕ_1 , by contrast, does vary: from 0.034 to 1.0 with median 0.30. It discriminates where the claim radius is constant, which is the plan’s argument for reporting it.

4.2 F2. The free rule is a sound lower bound with 16 exceptions

Against the committed radius on all 831 rows: **exact on 636**, within one on **813**, **over-certifying on 16**, under-certifying on 179. All 16 over-certifications have radius 1 and separation 2, and fall in five networks (`barley` 7, `win95pts` 4, `Schipf_2010` 2, `magic-niab` 2, `asia` 1).

Of the 179 under-certifications, 69 sit in the undefined stratum, where the rule predicts 1 because it has nothing to say; that stratum has 368 rows, of which 299 have radius 1 and 69 have radius 2. So the rule is exactly wrong on every undefined row whose radius exceeds 1, by construction.

[flag] The 16 exceptions are **not 16 independent counterexamples**. They arise from only **six distinct** (*network*, *X*) **pairs**: `barley/sorttkv` alone contributes 7 and `win95pts/AppDtGnTm` contributes 4. Given that the plan elsewhere argues carefully for network-level clustering (F7), the exception count should be reported at treatment level too, or the paper invites exactly the objection it is trying to pre-empt.

4.3 F3. The prior law transfers, but with counterexamples

The plan states the session-5 law as $r \geq \min(s, k_{G_0})$. On the 463 measurable rows: equality on 337, radius greater on 110, radius **smaller on 16**. The radius never exceeds k_{G_0} anywhere in the corpus, and equals it on 15 rows.

[flag] Sixteen rows violate a relation written as an inequality. I verified that the violating set is **identical** to the 16 over-certifying rows of F2 — the same six treatments. This is coherent, but the plan’s phrasing (“transfers as $r \geq \min(s, k_{G_0})$ ”) reads as a theorem when the data say it is an empirical regularity with a known, clustered exception set. Worth a word, since a reviewer

will check.

4.4 F4. The coverage sweep is not nested

The stride in `select_knowledge` selects indices $\lfloor i \cdot |K| / \text{keep} \rfloor$, which yields different index sets at different coverages. $K(0.5) \subseteq K(1.0)$ holds trivially on all 24 networks, but $K(0.25) \subseteq K(0.5)$ **fails on four**: diabetes (26, 13, 6), ecoli70 (10, 5, 2), magic-irri (10, 5, 2), munin1 (6, 3, 2).

The consequence is that the coverage sweep is not a monotone ablation. On the 105 pairs admissible at all three coverages, radius is unchanged on 73, lower at lower coverage on 32, and higher on none. And one pair — munin1, R_MED_RD_EW $\rightarrow$ R_MED_CV_EW — is admissible at coverage 1.0 and 0.25 but *not* at 0.5, which is only possible because the knowledge sets are not nested. The plan’s remedy (take prefixes of one seeded permutation) is the right fix and is scheduled.

4.5 F5. The 518 unidentified rejections are structurally unrecoverable

All **518** o_g0_not_identified rejections are **non-amenable** at G_0 . No adjustment set can be certified on them under *any* definition of the optimal set, so they are a genuine structural exclusion rather than a limitation of the current definition. This closes off a line of objection cheaply and is a good result to have in hand.

4.6 F6. Clustering and the real sample size

The intraclass correlation of the $r = 1$ indicator across 25 networks is **0.398** with mean group size $m_0 = 30.51$. The design effect is $1 + (m_0 - 1)\rho = 12.74$, giving an effective sample of $831/12.74 \approx 65$. The plan’s “about 60” is right, and its consequent decision — bootstrap over networks, print no per-row interval — follows.

5 Verification I performed

All checks were run read-only, from copies in a scratch directory; the working tree was clean before and after (`git status --porcelain` empty).

Check	Result
Re-ran <code>stage0_ladder.py</code> with <code>ROOT/OUT</code> redirected to scratch	[verified] <code>ladder_summary.json</code> and <code>ladder_rows.csv</code> reproduce byte-for-byte
Recomputed Check 1 from scratch (own script, same seven library calls) on mediator, asia, sachs, insurance, Shrier_2008, barley	[verified] 204 of 540 rows , zero mismatches in both <code>single_retractions_breaking_validity</code> and <code>n_asserted</code>
Recomputed Check 3 amenability on six networks	[verified] 255 of 518 rejections, zero amenable
Recomputed the free rule and all funnel counts directly from <code>instances.jsonl</code>	[verified] 636 / 16 / 179 and every per-network breakdown match
Diffed the script’s local <code>select_knowledge</code> against <code>benchmarks/measure.py:103</code>	[verified] Faithful copy: same round, step, and clamp

Check	Result
Traced the tautology claim to <code>fast_gate</code> at <code>measure.py:62</code>	[verified] Gate admission test $\equiv$ Check 1 at coverage 1.0
Checked internal arithmetic of both summaries (partitions, per-network sums)	[verified] All partitions sum correctly

I did not re-run `stage0_claim_radius.py` in full, since the six-network spot check (38% of rows, zero mismatches) establishes the computation, and a full run is slow.

6 Issues to raise before merge

None of these are computational errors. They are documentation and consistency defects; the first two are worth fixing before the abstract deadline.

6.1 I1. The Methods paragraph contradicts the plan (highest priority)

`paper/sections/evaluation-protocol.tex` is dated the same day as the plan but describes a different experiment.

1. It states that “for each network we estimate a CPDAG from one designated discovery run and draw treatment-outcome pairs from that estimate.” The plan’s design restricts estimated CPDAGs to a second panel on the **four Gaussian networks**, with the oracle CPDAG on the 25 yielding networks as the first panel. The file’s own header `TODO` flags exactly this, and the plan’s closing line acknowledges the sentence needs the two-panel clause. It is a known gap, but it is still an unresolved contradiction between two committed documents.
2. More seriously, the paragraph lists as one of four knowledge sources “knowledge read from each network’s own source publication, with a verbatim quotation required for every assertion.” The plan **explicitly defers this past the deadline** (“Deferred past the deadline: knowledge harvested from source publications”). As drafted, the Methods paragraph promises an arm that will not exist in the submission.

6.2 I2. A denominator mismatch resting on an inaccurate rationale

The two stage-0 artifacts use different populations: the ladder covers 831 rows including `pathfinder`, while the claim-radius run covers 540 and skips it. The stated justification — an 85-node component yielding “three censored rows” — is not what the data show.

The three records carrying `censored: true` are `_meta run-level` markers, one per coverage level, each recording a 1,200-second wall-clock timeout on the `pathfinder` sweep. `pathfinder`’s three *admissible* rows are “ok”, are **not** censored, and do appear in the ladder, where they contribute one under-certification.

The skip is defensible on cost grounds. But the real caveat is more important than the stated one and is currently recorded nowhere: `pathfinder`’s sweep was **truncated by a wall cap**, so its three admissible rows are a truncated sample of that network, not its full yield. The docstring should say that instead.

6.3 I3. Two small robustness gaps in `stage0_ladder.py`

- `paired_radius_across_coverage` classifies triples by comparing *only* 0.25 against 1.0; the 0.5 value never enters the comparison. Its three keys ($73 + 32 + 0 = 105$) happen to partition the triples today, but a pair whose radius dips only at 0.5 would fall through all three buckets silently. Given F4 — the sweep is known to be non-monotone — this is a live possibility once the prefix fix lands.
- `icc_one_way` divides by $k - 1$ and $n - k$ with no guard. A single-row network, or a run restricted to one network, raises `ZeroDivisionError` rather than reporting that the ICC is undefined.

6.4 I4. A duplicated definition

`stage0_claim_radius.py` re-implements `select_knowledge` rather than importing it. I verified the copy is currently faithful, but the plan schedules a change to exactly this function (nested coverage by prefix of one seeded permutation) for 12 September. The moment that lands, the copy becomes silently stale and the nesting finding becomes unreproducible. Import it, or assert equality against the library version at run time.

7 Project-level observations

Schedule. The plan is dated 11 September against a 25 September deadline with the abstract closing on the 18th. Stage 0 is the only completed item. The 13–16 September block carries the frame, the questionnaire, elicitation, the matched control, the radius sweep, the ladder on elicited knowledge, survival, cross-arm corruption, the audit, and the sampling band — ten workstreams in four days, all of them prerequisites for Tables 1–4. The five pipeline patches scheduled for the 12th are each a precondition for the rest, and three of them (nested coverage, the `not_amenable` split, the dispatch leg on every row) exist *because* of what stage 0 found.

Cross-branch dependency. The plan builds on three results it attributes to `survival-and-pareto` — efficiency invariance under truthful knowledge, correlated errors being more self-revealing than uniform ones ($1.7\times$ more silent failures for uniform), and strict dominance over SHD and $|K|$ holding on 7 of 9 strata and therefore *not* being claimed. It then asks that branch for three deliverables, including merging it into `main` so the paper reads from one branch. Merging this branch does not resolve that dependency, and the paper cannot be assembled until it is.

Intellectual quality. The strongest thing about this branch is that its author went looking for the weakness in his own instrument and found it. The decision to demote the hop radius to “geometry,” to price the certificate in the analyst’s own asserted claims, and to report the free rule as a lower bound with its exceptions rather than as a baseline to beat, all follow from evidence rather than from convenience — and all make the paper’s claims smaller and more defensible. The invariant stated in the design section (the true DAG draws the samples and audits the certificate afterwards, and is never on the computation path, with hashes on every row to enforce it) is the right discipline for this kind of work.

8 Recommendation

Merge. The branch adds no risk: it touches no source, its outputs reproduce exactly, and its findings survived independent re-derivation. Its central claim is negative and correct, and it is honest about being definitional.

Before the 18 September abstract deadline:

1. Reconcile `evaluation-protocol.tex` with the plan — the two-panel clause, and remove or explicitly defer the source-publication arm (I1).
2. Correct the `pathfinder` rationale to name the wall-cap truncation (I2).
3. Consider recording the atomic-perturbation clause as a status rather than a filter, to recover the 1,659 radius- ≥ 2 rows the gate currently discards (F1).
4. Report the free rule's 16 exceptions at treatment level as well as row level (F2).

The items in I3 and I4 can follow the 12 September patches, but I4 should not be left until after `select_knowledge` changes.

All figures in this review were re-derived from `results/axisa3/instances.jsonl` and the committed `results/stage0/` artifacts. No file in the repository was modified; verification ran from copies in a scratch directory.